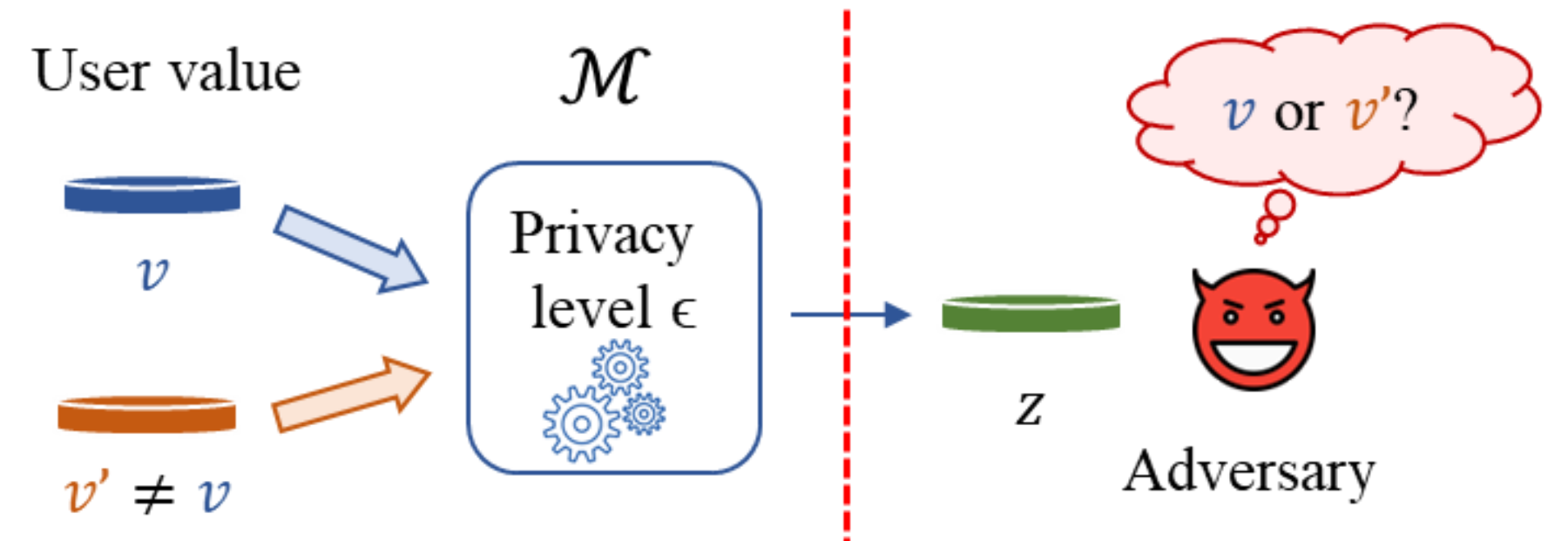


Revisiting Locally Differentially Private Protocols: Towards Better Trade-offs in Privacy, Utility, and Attack Resistance



Context & Motivation

- Local Differential Privacy (LDP) allows users to **privatize data before sharing**
- Widely deployed in **large-scale systems** (e.g., frequency monitoring in Big Tech)
- State-of-the-art LDP frequency estimation protocols are:**
 - Optimized for utility** (variance/MSE) [1]
 - Analyzed under reconstruction attacks (ASR)** [2,3]

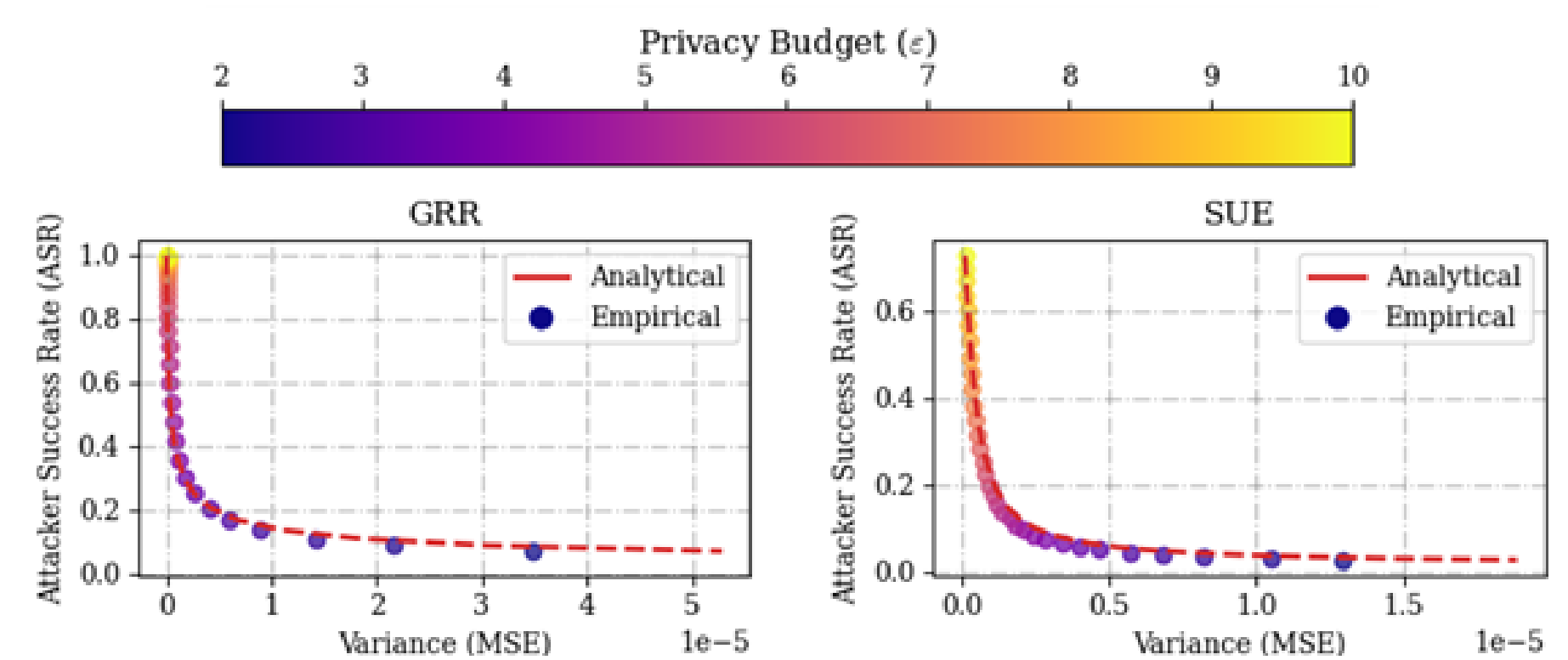


Gap

- No unified evaluation** of privacy-utility-attackability trade-offs

Our New Perspective for LDP Protocol Analysis

- Given ϵ , we **jointly** quantify:
 - Utility loss (MSE):** how far estimates deviate
 - Privacy risk (ASR):** reconstruction attack
- These define a **Pareto frontier** → optimal trade-offs
- Enables **principled comparison** of LDP protocols

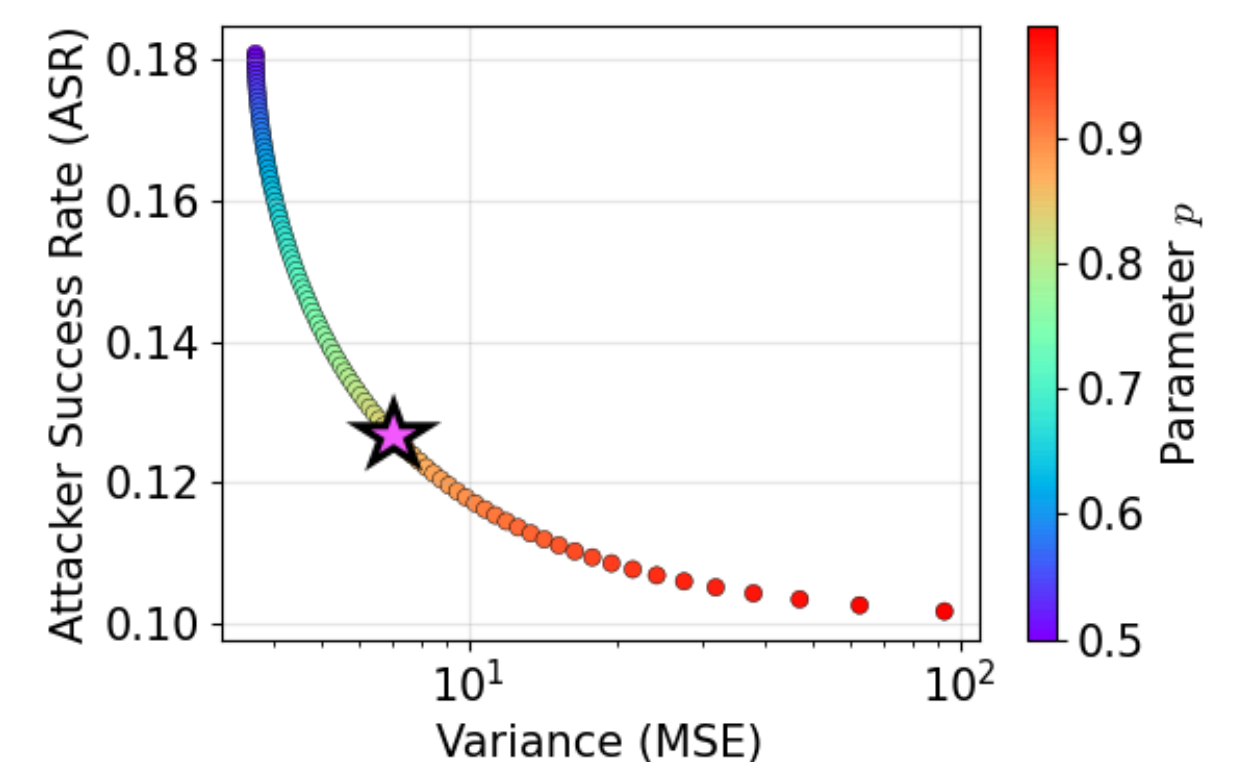


Ex: ASR-MSE trade-off (**analytical** vs. **empirical**)

From Analysis to Joint Optimization (ASR-MSE)

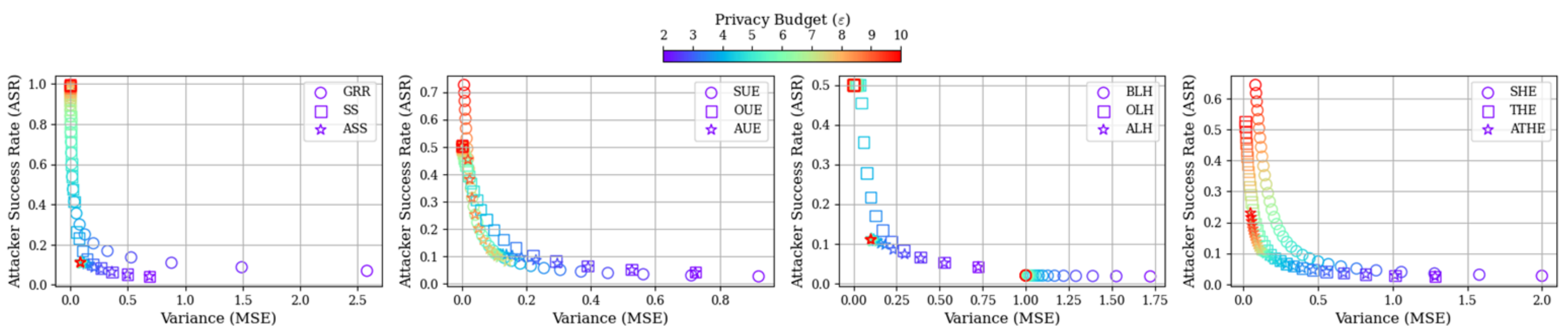
Insight: Some protocols have **tunable parameters** → we can **re-optimize them analytically**

- Approach:** For an LDP protocol **parameterized by p** :
 - Step 1:** Compute the Pareto frontier (ASR vs. MSE)
 - Step 2:** Select a point ★ with a **multi-objective rule** [4]
- ⇒ **Weighted Scalarization:**
$$\min_p J(p) = w_{ASR} \cdot ASR + w_{MSE} \cdot MSE$$
- Protocol design becomes a controllable trade-off, not a fixed choice**



Results & Takeaways

- Standard protocols become **adaptive** via parameter tuning (**ASS, AUE, ALH, ATHE**)
- Our adaptive protocols **strictly dominate** standard protocols on the Pareto frontier
- Our framework **naturally extends** to other risks (e.g., data poisoning)



References

- [1] Tianhao Wang, Jeremiah Blocki, Ninghui Li, and Somesh Jha. "Locally differentially private protocols for frequency estimation." USENIX Security 2017.
- [2] M. Emre Gursoy, et al. "An adversarial approach to protocol analysis and selection in local differential privacy." IEEE Transactions on Information Forensics and Security, 2022.
- [3] Héber H. Arcolezi, Sébastien Gambs, Jean-François Couchot, and Catuscia Palamidessi. "On the risks of collecting multidimensional data under local differential privacy." VLDB 2023.
- [4] Donoso, Yezid, and Ramon Fabregat. "Multi-objective optimization in computer networks using metaheuristics". Auerbach Publications, 2016.